

ICE-3107: Digital Signal Processing

Digital Filters

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Lecture Outcomes

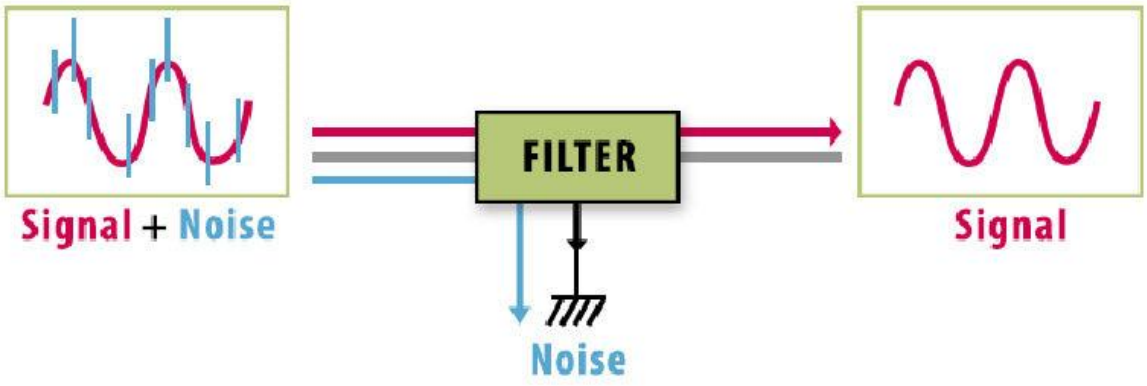
Understand the concept, properties, and applications of the digital filters.

Filter

A filter is a system that:

- Allows some frequencies to pass
- Blocks or attenuates other frequencies

Filters are widely employed in signal processing and communication systems in applications such as channel equalization, noise reduction, radar, audio processing, video processing, biomedical signal processing, and analysis of economic and financial data.



Filter (Cont.)

The primary functions of filters are one of the followings:

(a) To confine a signal into a prescribed frequency band as in low-pass, high-pass, and band-pass filters.

(b) To decompose a signal into two or more sub-bands as in filter-banks, graphic equalizers, sub-band coders, frequency multiplexers.

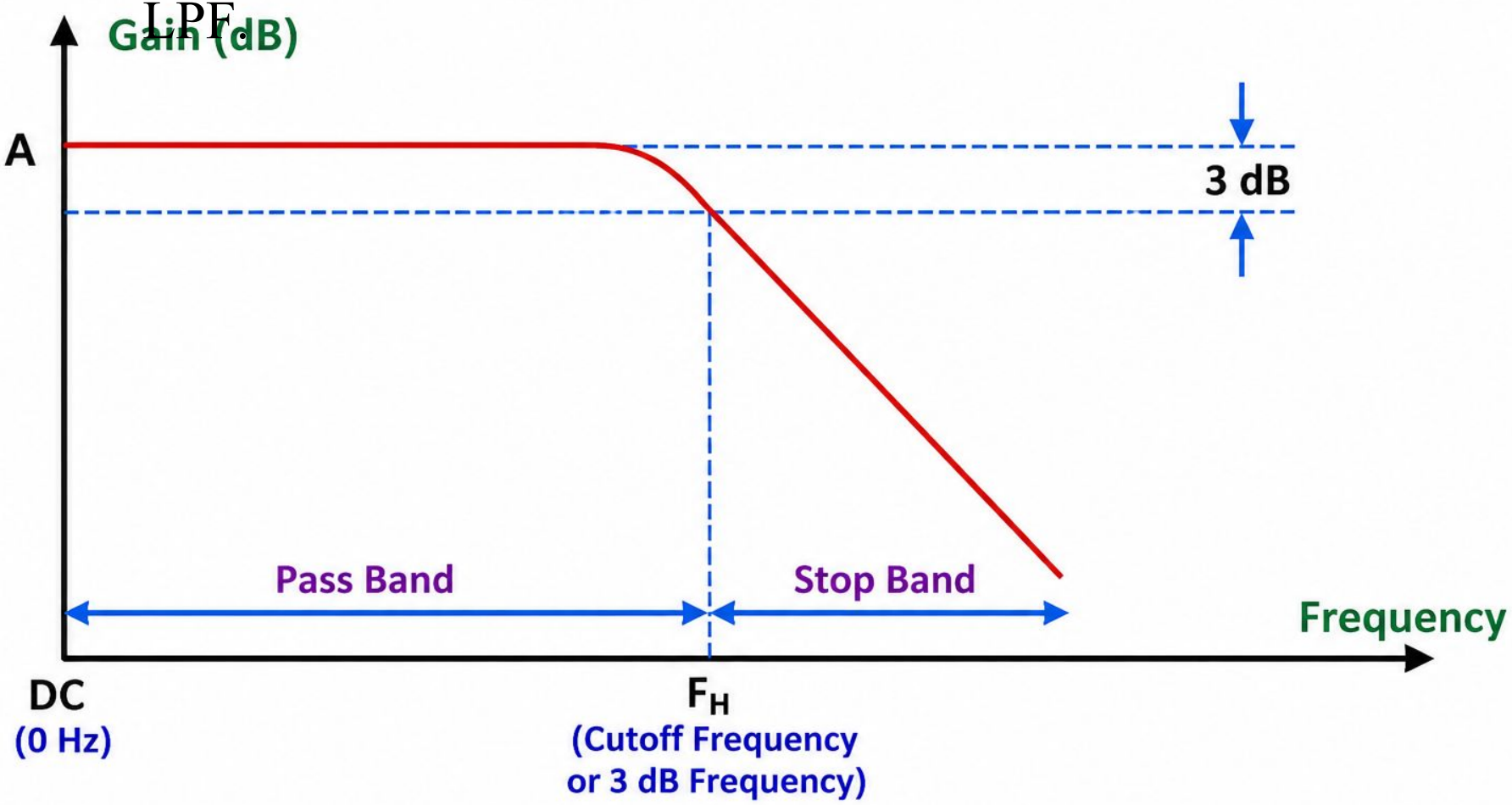
(c) To modify the frequency spectrum of a signal as in telephone channel equalization and audio graphic equalizers.

(d) To model the input-output relationship of a system such as

telecommunication channels, human vocal tract, and music synthesizers.

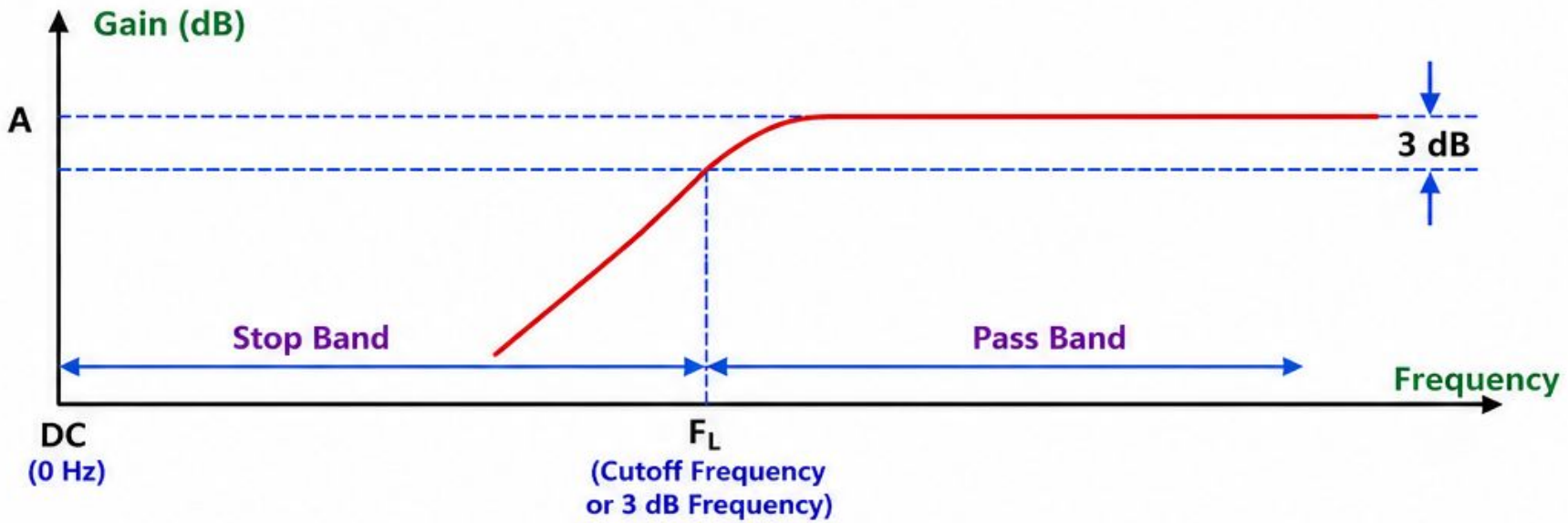
Low Pass Filter

- It passes low-frequency components through it and high-frequency components of the signal are blocked by LPF.



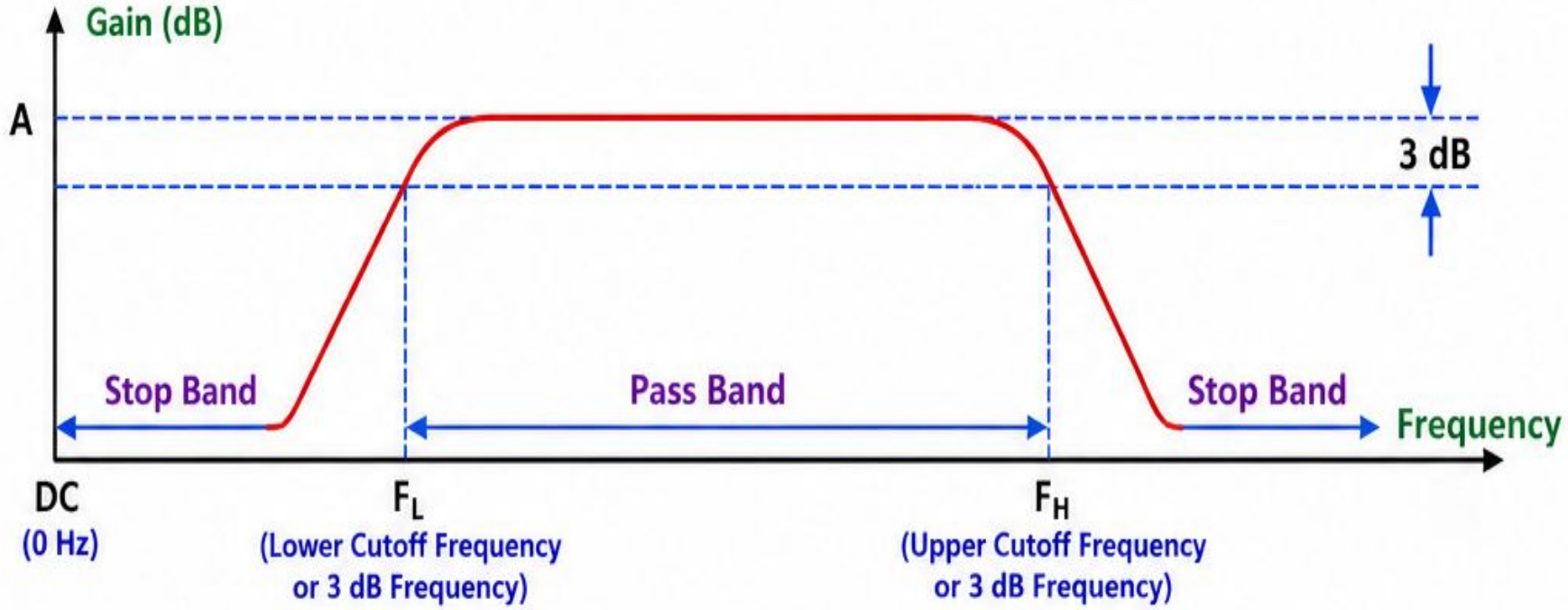
High Pass Filter

- It passes high-frequency components through it and low-frequency components of the signal are blocked by HPF.



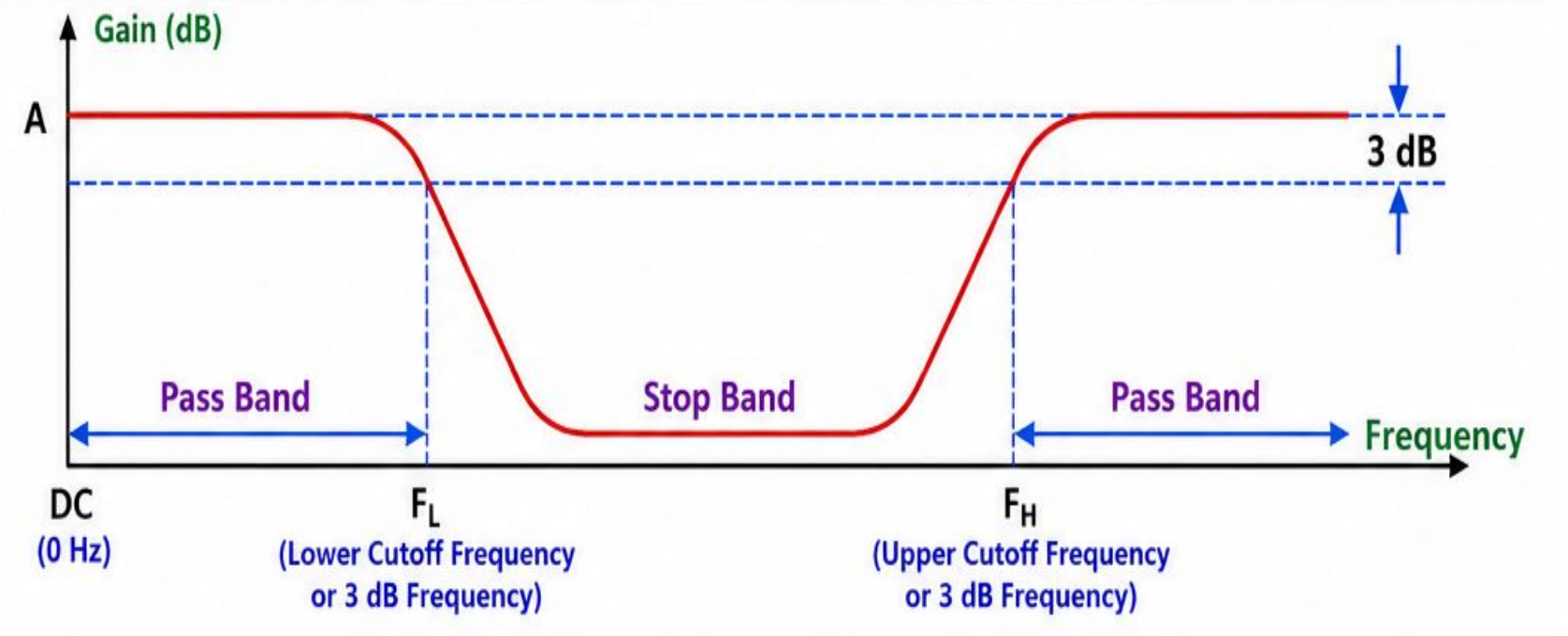
Band Pass Filter (BPF)

- It passes only specific range of frequencies.



Band Stop Filter (BSF)

- It blocks a specific range of frequencies.



Frequency Response Concept

- A filter does not treat all parts of a signal equally.
- Some frequencies are allowed to pass strongly, some are reduced, and some may be shifted in phase.

So, to understand a filter, we ask:

- What happens to low frequencies?
- What happens to high frequencies?
- How much are they reduced?
- Are they delayed or shifted?

That is exactly what frequency response tells us.

Frequency Response

The **frequency response** of a filter tells us how the filter behaves for each frequency.

For a digital filter, it is written as:

$$H\left(e^{j\omega}\right)$$

- H means the system or filter response.
- $e^{j\omega}$ represents a complex sinusoid at frequency ω.
- So $H\left(e^{j\omega}\right)$ tells us how the filter reacts when the input is a sinusoid of frequency ω.

Frequency Response (Cont.)

For an LTI system, the frequency response is defined as:

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

This means:

- $X(e^{j\omega})$ =frequency content of the input
- $Y(e^{j\omega})$ =frequency content of the output

Frequency Response (Cont.)

- If at some frequency ω,

$$| H(e^{j\omega}) | = 1$$

then that frequency passes unchanged in amplitude.

- If:

$$| H(e^{j\omega}) | < 1$$

then that frequency is attenuated.

- If:

$$| H(e^{j\omega}) | > 1$$

then that frequency is amplified.

Magnitude Response

The magnitude response is:

$$| H(e^{j\omega}) |$$

This is the absolute value, or magnitude, of the frequency response.

How strongly each frequency is passed or blocked.

- ✓ $| H(e^{j\omega}) | = 0.8$: The output amplitude at that frequency is 80% of the input amplitude.
- ✓ $| H(e^{j\omega}) | = 0.2$: Then only 20% remains.

Phase Response

The phase response is:

$$\angle H\left(e^{j\omega}\right)$$

- This tells us how much phase shift the filter introduces at each frequency.
- How much each frequency is delayed or shifted.

Why phase matters

Two filters may have the same magnitude response, but different phase response. That means they may reduce the same frequencies in the same way, but one may shift the signal more than the other.

This is important in:

•audio processing, communication, biomedical signals, and image processing

LPF & HPF

Example of a low-pass filter

For a low-pass filter:

- low frequencies: magnitude near 1
- high frequencies: magnitude near 0

Phase may change gradually depending on filter type.

So, the magnitude response shows why it is called low-pass.

Example of a high-pass filter

For a high-pass filter:

- low frequencies: magnitude near 0
- high frequencies: magnitude near 1

So, the magnitude response shows why it is called high-pass.

Magnitude Response:

$$| H_n(f) | = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2n}}}$$

where:

- n= filter order
- $f_c = 100\ Hz$

Frequency	1st Order Gain	2nd Order Gain	3rd Order Gain
80 Hz	0.781	0.842	0.889
100 Hz	0.707	0.707	0.707
120 Hz	0.640	0.570	0.503

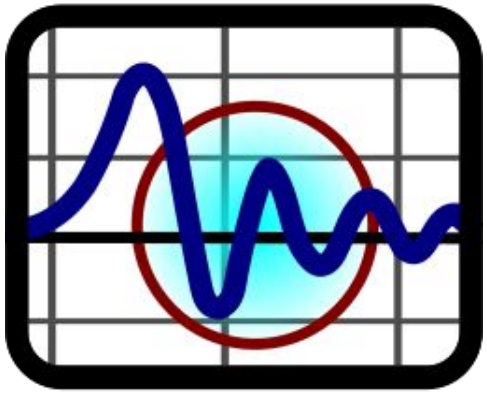
Frequency	1st Order	2nd Order	3rd Order
80 Hz	-2.15 dB	-1.49 dB	-1.02 dB
100 Hz	-3 dB	-3 dB	-3 dB
120 Hz	-3.87 dB	-4.88 dB	-5.97 dB

Digital Filter

A digital filter is an algorithm/system that processes discrete-time signals to remove or modify certain frequency components.

Digital filters are systems that process discrete-time signals to:

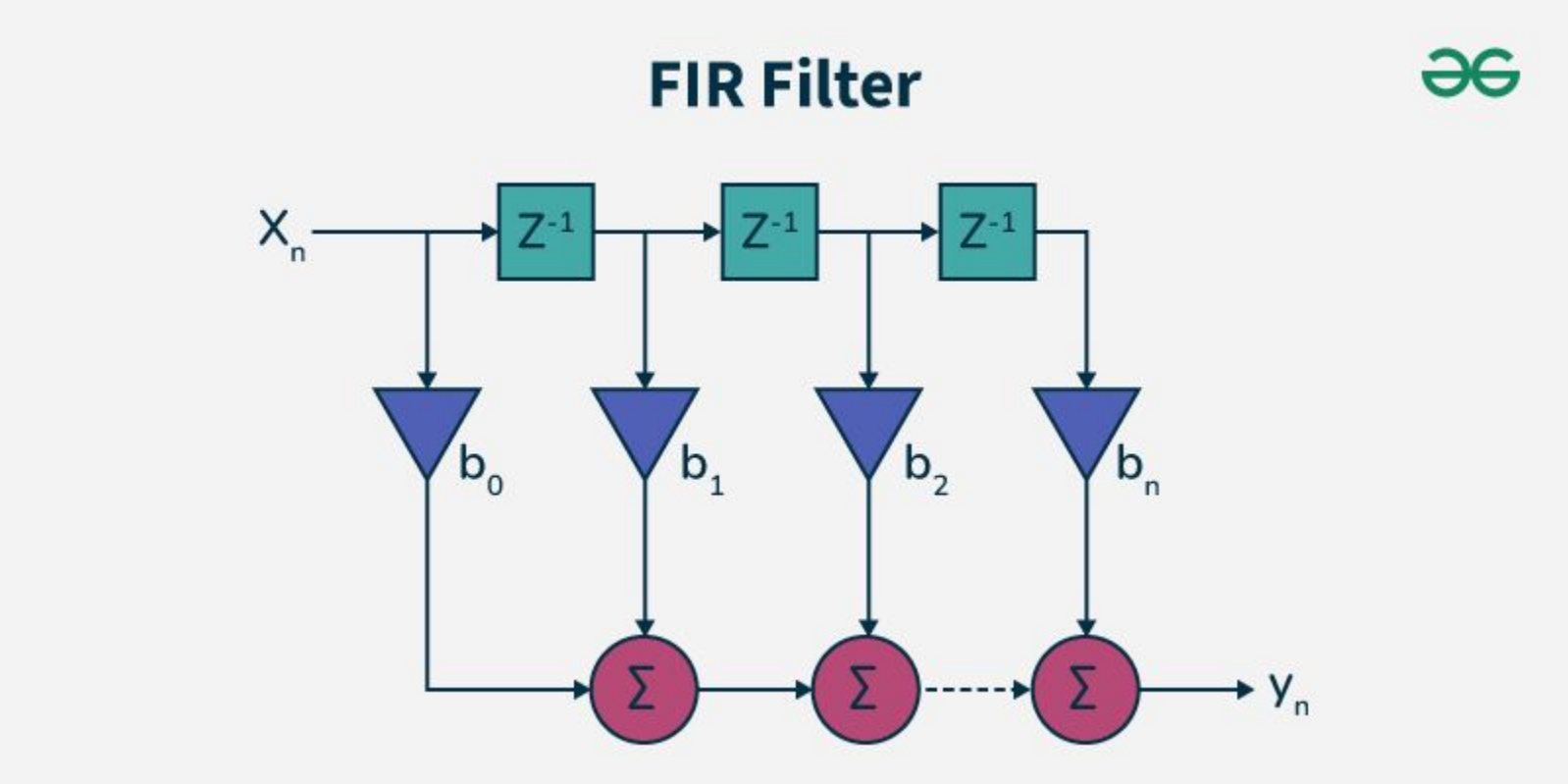
- remove noise,
- enhance useful information,
- separate frequency components,
- smooth or sharpen signals.



Finite Impulse Response (FIR) filter

A digital filter whose impulse response becomes zero after a finite number of samples.

FIR filters do not use feedback, meaning the output depends only on the present and past input values.



FIR filter (Cont.)

General FIR equation:

$$y[n] = \sum_{k=0}^{M} b_k\, x[n-k]$$

Expanded:

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] + \cdots + b_M x[n-M]$$

b_k =Filter coefficients

M =Filter order

Taking Z-transform:

$$Y(z) = b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z) + \cdots + b_M z^{-M} X(z)$$

Factor out $X(z)$:

$$Y(z) = X(z)(b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M})$$

FIR filter (Cont.)

Transfer Function:

$$H(z) = \frac{Y(z)}{X(z)}$$

$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}$$

or

$$H(z) = \sum_{k=0}^{M} b_k \, z^{-k}$$

Example FIR Filter

$$y[n] = x[n] + 0.5x[n-1]$$

FIR filter (Cont.)

Consider the FIR filter:

$$y[n] = x[n] + 0.5x[n-1] + 0.25x[n-2]$$

Suppose:

$$x[n] = \delta[n]$$

Since impulse input gives impulse response:

$$h[n] = y[n]$$

For $n = 0$, $h[0] = 1$
For $n = 1$, $h[1] = 0.5$
For $n = 2$, $h[2] = 0.25$
For $n = 3$, $h[3] = 0$
For $n = 4$, $h[4] = 0$

So, the response is: 1, 0.5, 0.25, 0, 0, 0,…

The impulse response becomes zero after a finite number of samples.

FIR filter (Cont.)

Practice Problem

Given FIR Filter:

$$y[n] = x[n] + 0.5x[n-1]$$

And input discrete-signal:

$$x[n]=\{1,2,3\}$$

Find output.

Assume initial past inputs are zero.

FIR filter (Cont.)

Solution

For $n = 0$

$$y[0] = 1 + 0 = 1$$

For $n = 1$

$$y[1] = 2 + 0.5(1) = 2.5$$

For $n = 2$

$$y[2] = 3 + 0.5(2) = 4$$

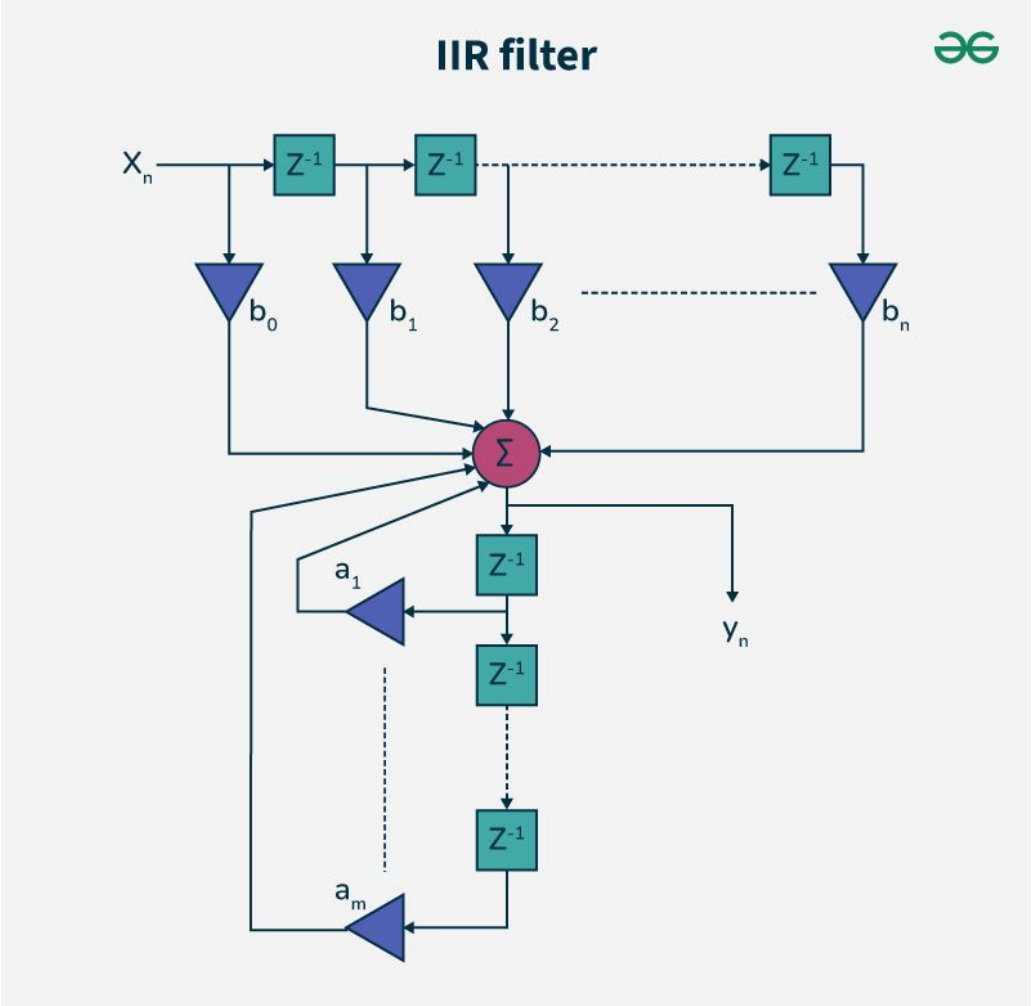
Output

$$y[n] = \{1, 2.5, 4\}$$

Infinite Impulse Response (IIR) filter

A digital filter whose impulse response theoretically continues indefinitely.

This occurs because IIR filters use feedback, meaning the output depends on both previous inputs and previous outputs.



IIR filter (Cont.)

General form:

$$y[n] = \sum_{k=0}^{M} b_k\, x[n-k] - \sum_{k=1}^{N} a_k\, y[n-k]$$

Expanded form:

$$y[n] = b_0 x[n] + b_1 x[n-1] + \cdots - a_1 y[n-1] - a_2 y[n-2]$$

- y[n−k] = Previous outputs
- a_k = Feedback coefficients

IIR filters have feedback

IIR filter (Cont.)

The **general transfer function of an IIR filter** is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

or expanded form:

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{1 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

IIR filter (Cont.)

$$y[n]=x[n]+0.5y[n-1]$$

Suppose:

$$x[n] = \delta[n]$$

For n=0, y[0]=1
For n=1, y[1]=0.5
For n=2, y[2]=0.25
For n=3, y[3]=0.125

Response

$$1,0.5,0.25,0.125, \ldots$$

continues forever.
Hence:
Infinite impulse response.

Realization of FIR Filter

- Derive the corresponding time-domain difference equation and draw the direct-form realization (block diagram) of the FIR filter.

FIR Transfer Function

$$H(z) = \sum_{k=0}^{N-1} h(k) z^{-k}$$

Substitute:

$$\frac{Y(z)}{X(z)} = \sum_{k=0}^{N-1} h(k) z^{-k}$$

Multiply both sides by $X(z)$:

$$Y(z) = X(z) \sum_{k=0}^{N-1} h(k) z^{-k}$$

Realization of FIR Filter (Cont.)

Using the Z-transform property

$$z^{-k}X(z) \leftrightarrow x(n-k)$$

Therefore

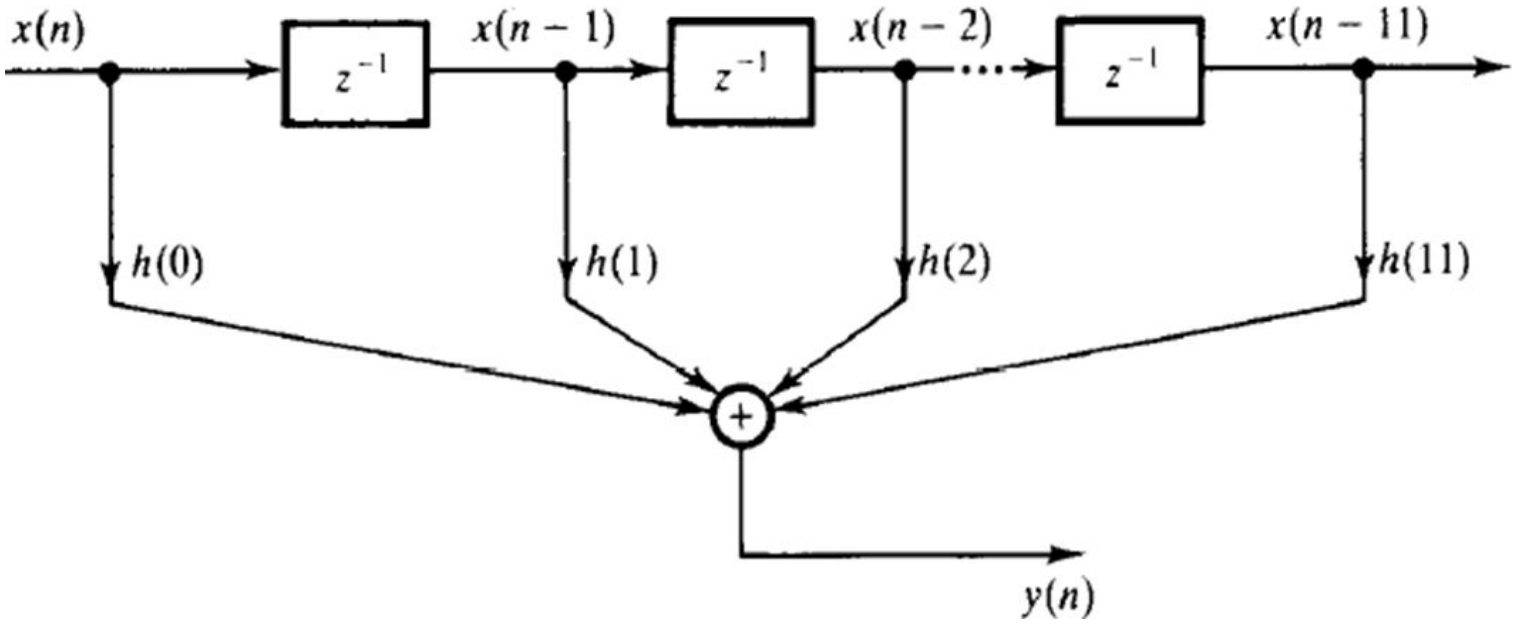
$$y(n) = \sum_{k=0}^{N-1} h\,(k)x(n-k)$$

Expand the Summation

$$y(n)$$
$$= h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \cdots$$
$$+ h(N-1)x(n-N+1)$$

Realization of FIR Filter (Cont.)

$$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \cdots + h(N-1)x(n-N+1)$$



Realization of a IIR Filter

Transfer Function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Decomposition Using an Intermediate Signal $\mathrm{V}(z)$

$$H(z) = \frac{Y(z)}{\mathrm{V}(z)} \cdot \frac{\mathrm{V}(z)}{X(z)}$$

Where:

$$\frac{\mathrm{V}(z)}{X(z)} = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

and

$$\frac{Y(z)}{\mathrm{V}(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2}$$

Realization of IIR Filter (Cont.)

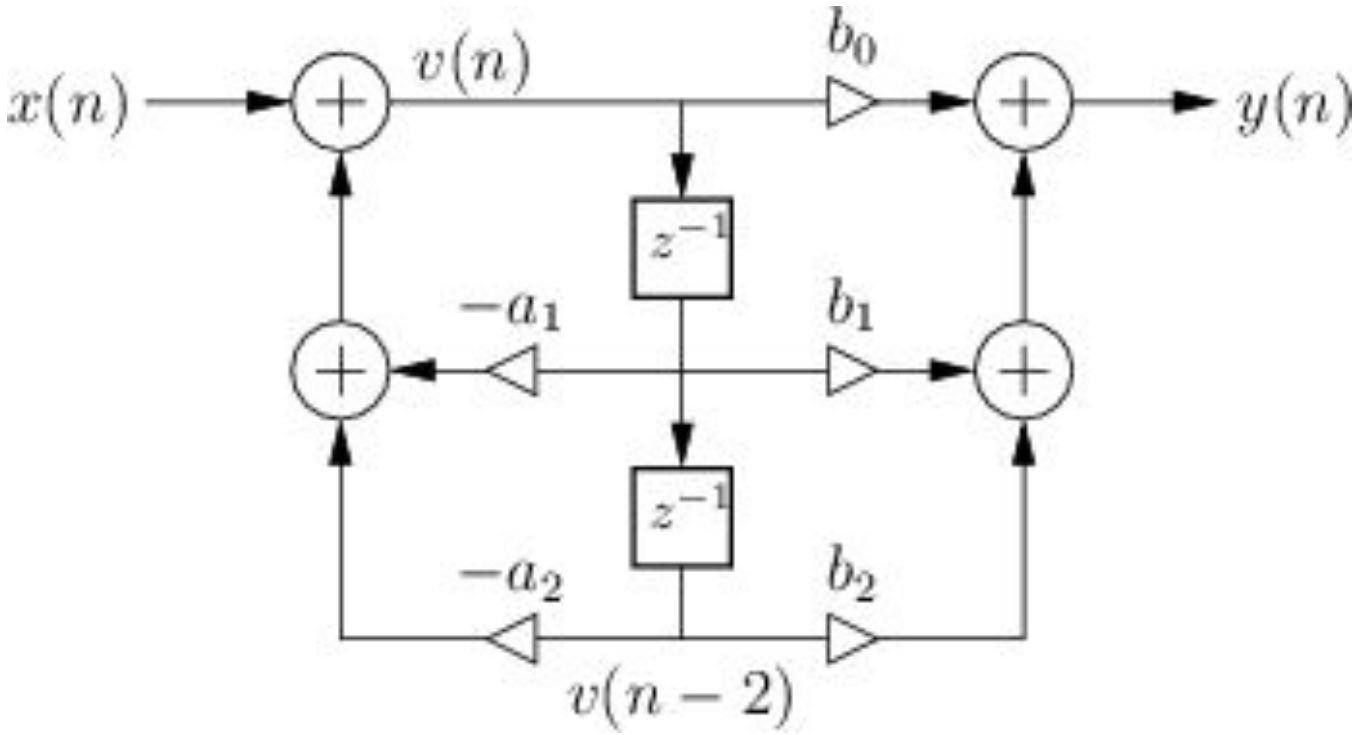
Difference Equations

Feedback (Recursive) Section

$$\mathrm{v}(n) = x(n) - a_1\mathrm{v}(n-1) - a_2\mathrm{v}(n-2)$$

Feedforward Section

$$y(n) = b_0\mathrm{v}(n) + b_1\mathrm{v}(n-1) + b_2\mathrm{v}(n-2)$$



Windowing Techniques in FIR Filter Design

One of the most popular methods for FIR filter design is the Window Method.

The basic idea is:

- Design an ideal filter in the frequency domain.
- Obtain its impulse response.
- Truncate the infinite impulse response using a window.
- Obtain a practical FIR filter.

Windowing Techniques (Cont.)

Windowing Concept

We multiply the ideal impulse response by a finite-length window:

$$h(n) = h_d(n)w(n)$$

Where:

- $h_d(n)$= ideal response
- $w(n)$= window
- $h(n)$= practical FIR response

Standard Window Profiles

❖ Rectangular Window

Simplest window.

$$w(n) = \begin{cases} 1, & 0 \le n \le N-1 \\ 0, & otherwise \end{cases}$$

❖ Hamming Window

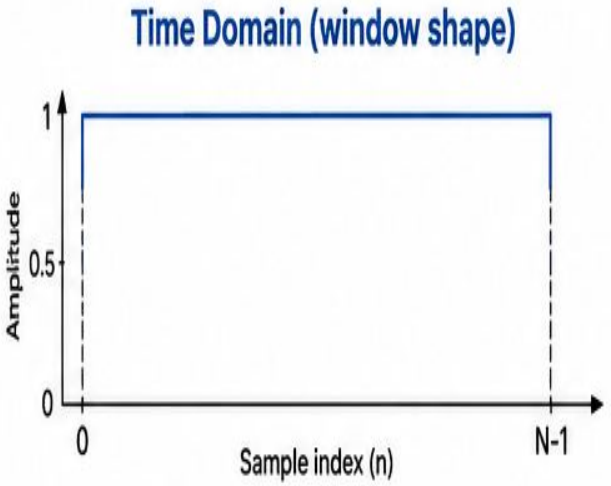
$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

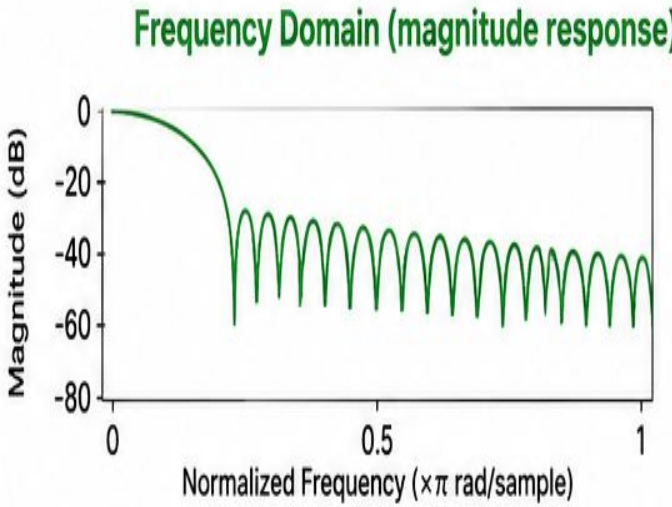
Standard Window Profiles

1) Rectangular Window

No tapering.

All coefficients are 1.





Highest ripple

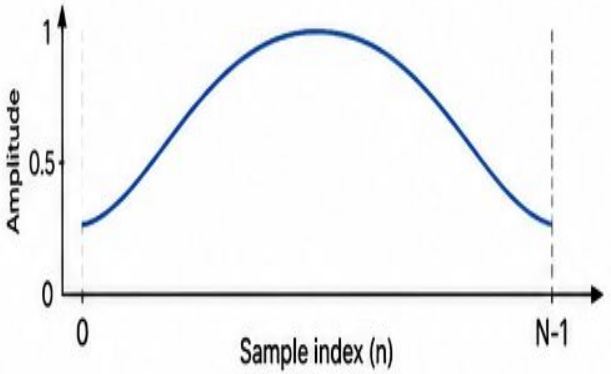
Poor stopband attenuation

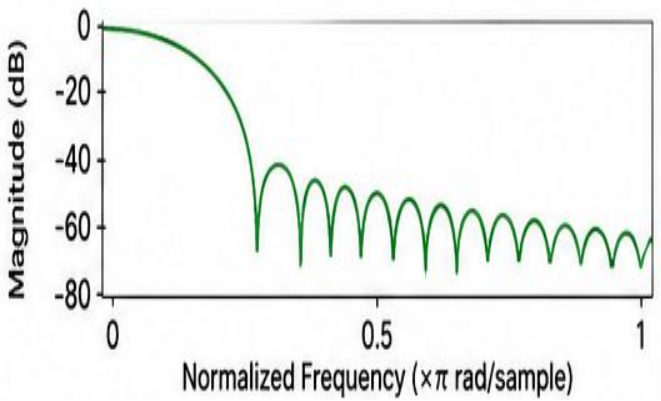
Narrowest transition band

2) Hamming Window

Tapers the ends.

Good balance.





Lower ripple than rectangular

Good stopband attenuation

Moderate transition band

FIR Design Procedure Using Window Method

Step 1: Specify filter requirements: Low-pass/High-pass/Band-pass /Band-stop

Step 2: Find ideal impulse response: $h_d(n)$

Step 3: Choose a suitable window

Step 4: Apply window

$$h(n) = h_d(n)w(n)$$

Step 5: Obtain FIR coefficients

$$h(0), h(1), \dots, h(N-1)$$

Step 6: Implement FIR filter

$$H(z) = \sum_{k=0}^{N-1} h\,(k)z^{-k}$$

Practice Problem 01

- Design a low-pass FIR filter using a rectangular window with:

$$N = 11$$

and cutoff frequency

$$\omega_c = \frac{\pi}{4}$$

Determine:

1. Ideal impulse response $h_d(n)$
2. Window coefficients $w(n)$
3. Actual FIR coefficients $h(n)$

Practice Problem 02

❖ **Hamming Window Coefficients**

For

$$N = 9$$

calculate the Hamming window coefficients using

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

for

$$n = 0,1,\ldots,8$$

2D Filters

A two-dimensional (2D) filter is a signal processing system that operates on data defined in two dimensions, such as digital images.

Unlike 1D filters that process signals varying with time, 2D filters process signals varying in both horizontal and vertical directions.

For a digital image f(x,y), a 2D filter modifies pixel values based on their neighboring pixels to achieve specific objectives such as noise removal, edge enhancement, smoothing, or feature extraction.

2D Filters (Cont.)

Mathematical Representation

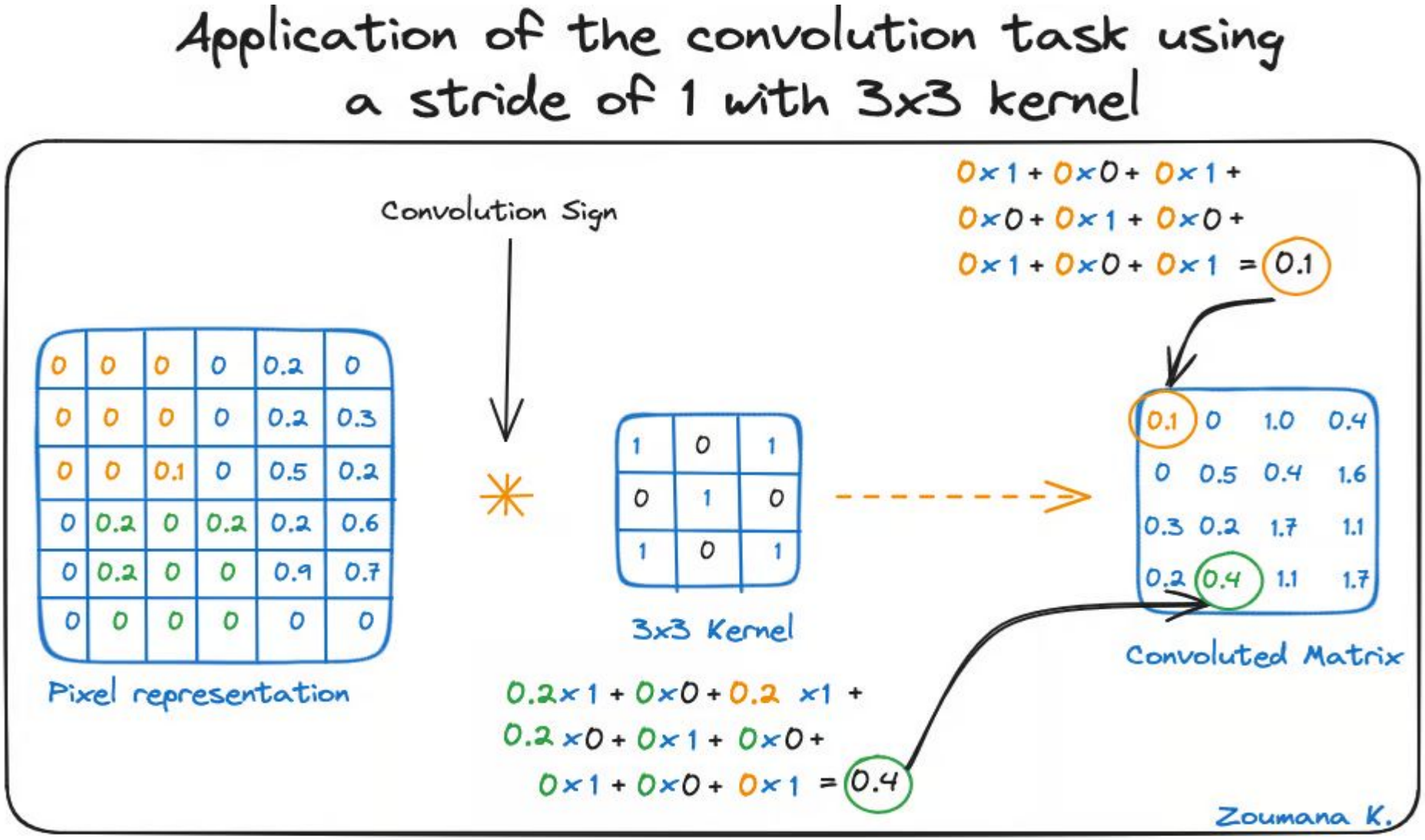
A 2D filter can be represented by a mask (kernel) $h(m, n)$. The output image $g(x, y)$ is obtained through 2D convolution:

$$g(x,y) = \sum_m \sum_n h(m,n) f(x-m, y-n)$$

where:

- $f(x, y)$ = input image
- $h(m, n)$ = filter kernel
- $g(x, y)$ = filtered image

2D Filters (Cont.)



Thank You


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